

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. Fibonacci

time limit per test: 0.25 seconds
memory limit per test: 256 megabytes

Find the n -th Fibonacci number modulo $10^9 + 7$.

So, you need to find F_n in the sequence defined as $F_0 = 0$, $F_1 = 1$ and $F_i = F_{i-1} + F_{i-2}$ for $i \geq 2$.

Input
An integer n ($0 \leq n \leq 10^{18}$).

Output
Print the answer modulo 1000000007.

input	Copy
3	
output	Copy
2	

input	Copy
6	
output	Copy
8	

input	Copy
50	
output	Copy
586268941	

Note
The first few terms of Fibonacci sequence are (0, 1, 1, 2, 3, 5, 8, 13, ...). In particular, we have $F_0 = 0$, $F_3 = 2$ and $F_6 = 8$. And for the last sample test:

$$F_{50} = 12586269025 \equiv 586268941 \pmod{10^9 + 7}$$

Matrix Exponentiation

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Submission	Time	Verdict
356010079	Dec/31/2025 22:02	Happy New Year!
335031259	Aug/22/2025 15:23	Happy New Year!
335031001	Aug/22/2025 15:21	Happy New Year!
328587418	Jul/12/2025 00:45	Happy New Year!
328587239	Jul/12/2025 00:42	Delivered to wrong address on flight 2 WA2
328587034	Jul/12/2025 00:38	Happy New Year!
328585216	Jul/12/2025 00:06	Delivered to wrong address on flight 1 WA1
328584343	Jul/11/2025 23:54	Happy New Year!
328572335	Jul/11/2025 21:49	Happy New Year!
328045308	Jul/08/2025 10:28	Happy New Year!